HEDGE FUND INVESTMENT RISK, RETURN, AND DIVERSIFICATION

Learning Outcome

- analyze sources of risk, return, and diversification among hedge fund investments

Hedge fund portfolios approach return generation differently from traditional portfolios, as Exhibit 5 shows. The most significant difference is that hedge funds seek to limit market exposure and returns from beta and primarily focus on generating idiosyncratic returns by identifying sources of unique return, or alpha. The primary source of hedge fund excess return is market inefficiencies (which may be short lived) and the skills of the manager in leveraging them.

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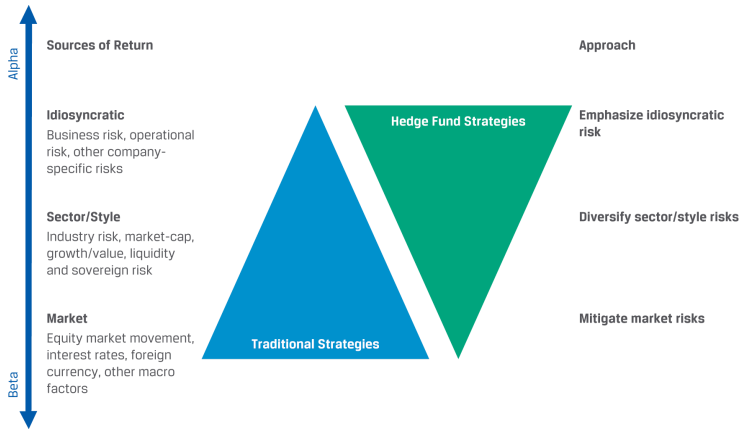
Exhibit 5:

Comparing Traditional and Hedge Fund Portfolio Sources of Return



Some specific sources of alpha are the manager skills in specific stock selection and utilizing higher-return strategies that minimize risks. The performance of hedge funds can be attributed to three distinct sources:

- Market beta—the broad market beta that can be realized using market index–based funds/ETFs
- Strategy beta—the beta attributed to the investment strategy of the hedge fund applied across the broad market
- Alpha—the manager-specific returns, due to the selection of specific positions

The managers can realize the strategy beta and alpha returns due to their skills in identifying mis-priced securities and sectors, correctly timing the market, and utilizing operational control on the company business model, as well as using leverage to amplify the results.

This does not suggest that the traditional asset pricing models are incorrect. They operate within a set of assumptions—not the least of which is that the markets are efficient. The returns from hedge funds are due to systematic and idiosyncratic alternative risk factors not considered by the model.

underperforming hedge funds that close out for performance issues face another problem. The capital redeemed from liquidated positions may result in a lower payout, effectively diminishing the total return from the fund.

To compare the risk-adjusted return characteristics of individual hedge funds and aggregate hedge fund strategies, we use hedge fund indexes that are created using publicly available hedge fund performance data. There are several vendors that create such indexes, and each index has its own features, but there are some general considerations around these indexes that distinguish them from other indexes used for performance comparison. Most hedge fund indexes are based on information reported by the hedge fund managers and others who receive performance information with the right to share that information. It should also be noted that the reporting by any hedge fund is voluntary. This situation introduces several sources of bias and suggests that hedge fund performance is likely overestimated.

Selection bias can be an issue. Individual funds are allocated to strategy peer groups in an inconsistent manner: sometimes based on the prospectus, sometimes based on historical style analysis, sometimes using a combination of approaches, each of which adds and compounds bias. Indexes may also have inconsistent sources of the underlying data. A handful of these indexes operate with transparent and constant inclusion, selection, and exclusion of individual managers and/or their funds. Some index providers impose requirements on AUM, vintage, and whether the hedge fund is open to new investors.

Survivorship bias can also be an issue. When funds that have stopped reporting are removed from the index, the index will likely show better performance. This effect can be mitigated by including the returns of funds of funds that are active and those that have stopped reporting. For instance, a hedge fund manager consistently underperforming a peer group may have limited incentives to continue to highlight underperformance, particularly if the manager is actively sourcing new investors or launching new funds. Hence, the manager may discontinue reporting poor performance, which overestimates reported performance.

Additionally, hedge funds that are closed to new investors or closed because of underperformance are treated equally. Their performance is excluded from the index value. Additionally, the hedge fund performance data are published with a delay, normally a time lag of four weeks or one month. Since these indexes are non-investable and illiquid, replicating their performance may be difficult.

In addition, backfill bias can be an issue. When a successful fund starts reporting performance for the first time, it is very likely that its past performance was stellar, since typically only hedge funds with favorable returns are reported. When such funds are added to an index, the benchmarks will overstate the actual performance. This is a variation of the survivorship bias that results when a new hedge fund is included in a given index and its past performance is “backfilled” into the index’s database. Typically, larger indexes have less backfill bias.

A large majority of hedge fund indexes are not weighted by assets under management. In these hedge fund indexes, each hedge fund receives an equal index weighting in the performance peer group. As a result, comparison of large and small funds can be skewed compared to the performance of a size-weighted index.

Hedge Fund Investment Risks and Returns

Many traditional investment funds, such as long-only mutual funds and index ETFs, diversify away much of the idiosyncratic risks in their holdings by investing in a large number of stocks and achieve most of their returns primarily by bearing the systematic market risk (beta). Unlike in traditional funds, hedge funds use a variety of instruments across asset classes and techniques in seeking to generate absolute returns in all market environments.

the minimal level of disclosure (as well as the frequency of disclosure) they offer, it is hard to conduct any performance attribution analysis. Additionally, the relative illiquidity of investments held by the funds makes marking to market a problematic and potentially futile process.

Hence, attributing the sources of returns and risks is a composite process that is further complicated by the complexity of the strategies and compounded using various sources of leverage. That is why risk and return comparisons are typically made to fund-of-funds composite indexes to minimize return distortions. That approach controls for the effects of self-reporting and selection biases. Furthermore, it ensures that the fund-of-funds benchmark index is investable.

Nonetheless, over the long term (since 1990), hedge funds have enjoyed higher returns (at least prior to fees and expenses) than either stocks or bonds and a standard deviation nearly identical to that of bonds. Specifically, Exhibit 6 compares the returns, risk, and performance measures for the HFRI Fund of Funds Composite Index, the MSCI ACWI Index, and the Bloomberg Barclays Global Aggregate Index. The HFRI Fund of Funds Composite Index is an equally weighted performance index of funds of hedge funds included in the HFR Database.

As mentioned earlier, hedge fund indexes typically build on data that each hedge fund reports. Because these data are self-reported, there is a degree of survivorship bias that the HFRI Fund of Funds Composite Index mitigates by reflecting the actual performance of portfolios of hedge funds. The measures shown here may reflect a lower reported return because of the added layer of fees, but they likely represent a fairer, more conservative, and more accurate estimate of average hedge fund performance than HFR's composite index of individual funds. The returns are likely biased toward equity long/short funds since these are frequently a substantial portion of funds of funds' allocation mix. There are also other hedge fund indexes, allowing a more thorough analysis of the source of returns.

As shown in Exhibit 6, over the 25-year period between 1990 and 2014, hedge funds had higher returns than either stocks or bonds and a standard deviation almost identical to that of bonds. This, however, depends on the benchmarks used, which typically are broad market benchmarks. In the 1990- 2014 period, hedge funds had modest overall correlation with global stock returns (0.56) and negligible correlation with global bond returns (0.07) over this 25-year period. Hedge funds certainly offered added value to institutional investors as a portfolio diversification agent in this time period.

Exhibit 6:

Historical Risk–Return Characteristics of Hedge Funds and Other Investments, 1990–2019

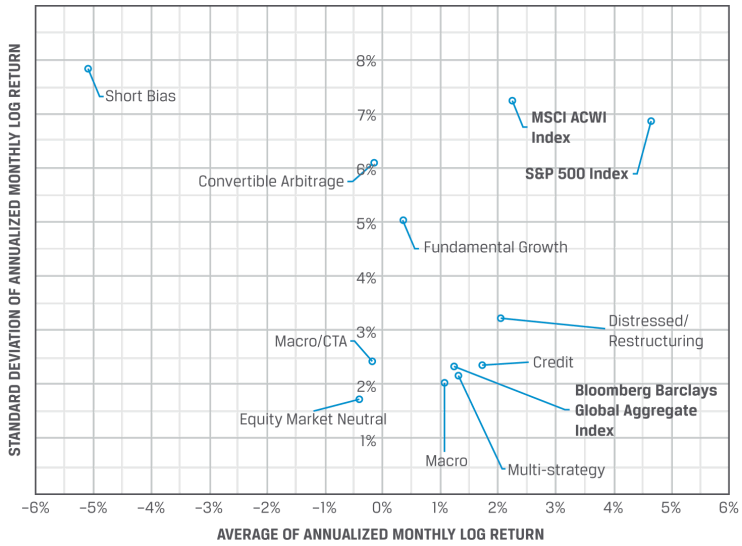
	FoF	Global Stocks	Global Bonds
Annualized return			
1990–2014	7.2%	6.9%	6.3%
2015–2019	*2.48%*	*10.41%*	*2.42%*
Annualized volatility			
1990-2014	6.0%	16.5%	5.8%
2015–2019	*1.08%*	*11.42%*	*1.33%*
FoF correlation (avg. monthly)			
1990–2014		0.56	0.07
2015–2019		*0.86*	*0.03*
Percentage of positive months	69.3%	61.3%	62.7%
Best month	6.8%	11.9%	6.2%
Worst month	−7.5%	−19.8%	−3.8%
Worst drawdown	−22.2%	−54.6%	−10.1%

Notably, Exhibit 6 shows that for the subsequent five-year period between 2015 and 2019, the absolute return of funds of funds relative, in particular, to global equities has declined, while their performance correlation with equity markets actually increased. This trend has made hedge fund allocations arguably less useful and also somewhat less popular. But some allocators have continued to find value to maintain or actually increase their allocation to a mix of hedge funds as a bond market substitute in their overall portfolio building.

Looking at the performance of various hedge fund strategies over time, an approach is to examine the relationship between hedge fund returns and the risk measured by the standard deviation of returns. Exhibit 7 shows the log returns and standard deviation of returns for selected hedge fund strategies on an annualized monthly basis as reported by Hedge Fund Research. The data cover the period from January 2008 to November 2021, a period of almost 14 years.

Exhibit 7:

Return and Risk of Selected Hedge Fund Strategies, 2008–2021



Source: Hedge Fund Research (HFR), annualized monthly log returns.

Some of the hedge fund strategies generated negative returns on average. Short-bias strategies underperformed all the other strategies in terms of both return and risk measured by standard deviation. For comparison, the S&P 500, MSCI ACWI, and Bloomberg Barclays Global Aggregate Index returns are also plotted to show that many strategies have widely varying risk/return characteristics compared to common equity and fixed income benchmarks.

Another approach to analyze the benefits from investing in hedge funds looks at the risk/return trade-off measured by the coefficient of variation of annual hedge fund returns. The coefficient of variation can be thought as the price of return in terms of risk or the relative return adjusted for risk: A higher coefficient of variation provides greater return for the same amount of risk.

Diversification Benefits of Hedge Fund Investments

The original hedge fund strategy was a market-neutral strategy in which the long and short positions made the portfolio beta neutral and generated a positive alpha. Stated differently, hedge funds initially kept their portfolios' net market sensitivity exposure to the market benchmark index at zero. By holding long and short positions, they sought to create value for their investors with minimal exposure to the market.

While hedge fund strategies vary widely, market-neutral, relative value, and event-driven strategies tend to outperform equity markets during market downturns and when individual stock correlations fall and tend to have weaker performance when correlations are high (reducing relative value opportunities) and equities move uniformly higher.

The diversification benefits of hedge funds first came to prominence with the dot-com bubble unwinding in 2000–2002, when they generally performed well compared with traditional long-only investment products. Starting with funds of funds, institutional investors increased their hedge fund exposure and then expanded it through direct allocations following the 2008 financial crash. Seeking better diversification and risk mitigation, despite high hedge fund fees, the investors sought absolute and uncorrelated risk-adjusted returns rather than outsized upside performance. In the early 2000s, when hedge funds came to prominence, the primary drivers were the absolute excess returns, regardless of market direction. The verification efforts from investors on hedge fund operations were minimal. The 2008 crisis (including Lehman Brothers' bankruptcy and the Bernie Madoff fraud) brought to light the need for additional due diligence. The correction after the 2008 crisis brought several changes. The inflow from high-quality institutional investors and pension funds increased. Such investors demanded additional transparency and control over the fund's investment process, fund operations, leverage applied, and managers' performance. This change has benefited only the highest-quality funds, and most other hedge funds failed to keep pace with the positive equity and bond market advances of 2009–2019. They continue to have a place in institutional asset allocations because of their risk-diversification properties.

Exhibit 8 shows return relationships between hedge funds and US stock and bond indexes from 2001 to 2021. Although diversification is typically used to reduce idiosyncratic risk, diversification using hedge fund investments seeks to add idiosyncratic outperformance.

While hedge fund risk diversification benefits can merit investigation, experience also reflects that their prospective advantage can vary over time. Investors need to be thorough in conducting due diligence when selecting a hedge fund manager. Hedge fund performance has a very low correlation with that of traditional asset classes, such as investment bonds and currencies/cash. For traditional investors in these asset classes, hedge funds provide an enhanced opportunity for portfolio diversification and for generating consistent returns over time. Adding hedge funds to a traditional 60/40 portfolio typically decreases the total portfolio standard deviation and increases the Sharpe ratio, thus enhancing portfolio diversification and risk-adjusted return.

Exhibit 8:

Historical Hedge Fund Monthly Return Correlations, 2001–2021

	Hedge Funds	S&P 500	Investment-Grade Bonds
Hedge Funds	1	0.82	0.10
S&P 500		1	−0.06
Investment-Grade Bonds			1

Notes: Hedge funds = Dow Jones Credit Suisse Hedge Fund Index. Investment-grade bonds = Bloomberg Barclays US Aggregate Bond Index.

As Exhibit 8 shows, hedge funds have a fairly high correlation with equities and a low correlation with investment-grade bonds.

Question Set

with the management of which type of fund?

A. Equity index ETF
B. Long-only equity mutual fund
C. Long/short equity hedge fund

Hide Solution

Solution:

C is correct. Long/short equity hedge funds are the most likely type of funds to seek out trades with high idiosyncratic risk, so their performance is driven primarily by stock selection and less by other factors, such as sector and market exposures. B is incorrect because long-only mutual funds are more likely to seek returns from market risk and sector risks with less emphasis on undiversifiable, security-specific risks. A is incorrect because equity index funds do not seek out idiosyncratic risk and simply try to mimic the performance of an index.

2. Hedge funds are not obligated to report their performance to the public. Many do, however, to attract additional investors. Hedge fund benchmarks and indexes are created using the reported data. Which of the following is *least likely* a bias that hedge fund investors should be aware of when evaluating the performance of hedge fund benchmarks and indexes?

A. Survivorship bias: Unsuccessful and liquidated hedge funds are removed from an index, resulting in their underperformance not being accounted for in the index.
B. Self-reporting bias: Hedge funds have the freedom to not report their performance, and only the ones that report are included in an index.
C. Societal bias: This relates to the perception of hedge funds at a societal level, where hedge funds that are perceived to be high performing are included in the index.

Hide Solution

Solution:

C is correct. Societal bias is not one of the biases that hedge fund investors need to be aware of when evaluating hedge fund performance. A and B are biases that distort the performance of hedge fund benchmarks and indexes.

3. An investor wants to invest in a diversified hedge fund that minimizes the return correlation with the traditional asset classes but would prefer the fund to be more liquid and transparent while minimizing the leverage obtained by borrowing or shorting. What would be the *most* appropriate hedge fund the investor can choose?

A. Fundamental value
B. Managed futures
C. Multi-strategy
D. Fund of funds

Hide Solution

Solution:

B is correct. Managed futures have, historically, exhibited low correlation with traditional assets and invest in active futures in liquid commodities and foreign exchange markets. They are also able to increase exposure without resorting to borrowing or shorting.

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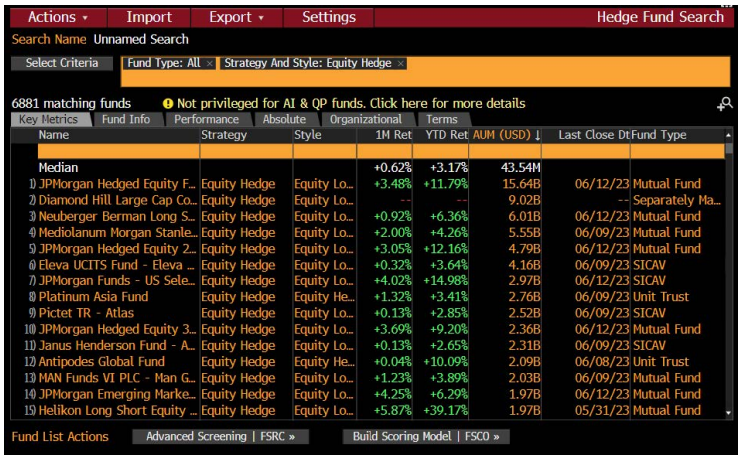

Name	Strategy	Style	1M Ret	YTD Ret	AUM (USD) 1	Last Close Dt	Fund Type
Median			+0.62%	+3.17%	43.54M		
1) JPMorgan Hedged Equity F...	Equity Hedge	Equity Lo...	+3.48%	+11.79%	15.64B	06/12/23	Mutual Fund
2) Diamond Hill Large Cap Co...	Equity Hedge	Equity Lo...	--	--	9.02B	--	Separately Ma...
3) Neuberger Berman Long S...	Equity Hedge	Equity Lo...	+0.92%	+6.36%	6.01B	06/12/23	Mutual Fund
4) Mediolanum Morgan Stanle...	Equity Hedge	Equity Lo...	+2.00%	+4.26%	5.55B	06/09/23	Mutual Fund
5) JPMorgan Hedged Equity 2...	Equity Hedge	Equity Lo...	+3.05%	+12.16%	4.79B	06/12/23	Mutual Fund
6) Eleva UCITS Fund - Eleva ...	Equity Hedge	Equity Lo...	+0.32%	+3.64%	4.16B	06/09/23	SICAV
7) JPMorgan Funds - US Sele...	Equity Hedge	Equity Lo...	+4.02%	+14.98%	2.97B	06/12/23	SICAV
8) Platinum Asia Fund	Equity Hedge	Equity He...	+1.32%	+3.41%	2.76B	06/09/23	Unit Trust
9) Pictet TR - Atlas	Equity Hedge	Equity Lo...	+0.13%	+2.85%	2.52B	06/09/23	SICAV
10) JPMorgan Hedged Equity 3...	Equity Hedge	Equity Lo...	+3.69%	+9.20%	2.36B	06/12/23	Mutual Fund
11) Janus Henderson Fund - A...	Equity Hedge	Equity Lo...	+0.13%	+2.65%	2.31B	06/09/23	SICAV
12) Antipodes Global Fund	Equity Hedge	Equity He...	+0.04%	+10.09%	2.09B	06/08/23	Unit Trust
13) MAN Funds VI PLC - Man G...	Equity Hedge	Equity Lo...	+1.23%	+3.89%	2.03B	06/09/23	Mutual Fund
14) JPMorgan Emerging Marke...	Equity Hedge	Equity Lo...	+4.25%	+6.29%	1.97B	06/12/23	Mutual Fund
15) Helikon Long Short Equity ...	Equity Hedge	Equity Lo...	+5.87%	+39.17%	1.97B	05/31/23	Mutual Fund

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The coefficient of variation can be thought as the price of return in terms of risk or the relative return adjusted for risk: A higher coefficient of variation provides greater return for the same amount of risk.

But I thought it was the other way, like lower COV provides greater returns for same amount of risk

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